An Extension of the Baire Property

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Outline

1. Preliminaries
2. Residually Null Sets
3. Extended Baire Property
4. Martin's Axiom
5. Future Work
6. References

Preliminaries

Definition

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Let X be a topological space. We say that $A \subseteq X$ is **nowhere dense** if $\mathrm{int}(\mathrm{cl}(A)) = \emptyset$ and that it is **meager** if there exists $\{B_n \subseteq X : n \in \mathbb{N}\}$ so that each B_n is nowhere dense and $A \subseteq \bigcup\{B_n : n \in \mathbb{N}\}$. A set $A \subseteq X$ is **co-meager** if $X \setminus A$ is meager.

- $\mathcal{M}(X)$ – the collection of all Borel probability measures on X
- $C(X)$ – the collection of continuous functions $f : X \to \mathbb{R}$
- $\mu \in \mathcal{M}(X)$ given induces a natural map $f \mapsto \int f \, d\mu$, $C(X) \to \mathbb{R}$
- $\mathcal{M}(X)$ injects naturally into $\mathbb{R}^{C(X)}$ so inherits its topology, called the **weak topology** therefrom

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Theorem

If X is Polish, then $\mathcal{M}(X)$ endowed with the weak topology is a Polish space.

Definition

A σ**-ideal** of sets is a collection $I \subseteq \wp(X)$ so that

- if $A \in I$ and $B \subseteq A$, then $B \in I$ and
- for $\{A_n : n \in \mathbb{N}\} \subseteq I$, $\bigcup\{A_n : n \in \mathbb{N}\} \in I$.

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By construction, the class of meager subsets of a topological space X is a σ-ideal.

Theorem (Baire Category Theorem)

The σ-ideal of meager subsets of a Polish space X is proper. That is, $X \notin I$.

Residually Null Sets

Given $\mu \in \mathcal{M}(X)$, recall that the outer measure μ^* induced by μ is a set function $\mu^* : \wp(X) \to [0,1]$ defined by

$$\mu^*(A) = \inf\{\mu(U) : U \text{ is open and } A \subseteq U\}.$$

Definition

For $A \subseteq X$, we define

$$\mathcal{N}(A) = \{\mu \in \mathcal{M}(X) : \mu^*(A) = 0\}.$$

Then we say that $A \subseteq X$ is **residually null** if $\mathcal{N}(A)$ is co-meager in $\mathcal{M}(X)$. Let $\mathbf{RN}(X)$ be the collection of residually null sets.

The following was done by Dubins and Freedman [1] in 1964 for the context when X is compact metrizable.

Theorem

Suppose X is a Polish space. If $A \subseteq X$ is meager, A is residually null. On the other hand, if A is a set with the Baire property and is residually null, then A is meager.

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- CH, or MA, $\Longrightarrow$ existence of residually null sets which are non-meager.

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Theorem

If X is a Polish space, then $\mathcal{M}_c(X)$ is a dense G_δ subset of $\mathcal{M}(X)$.

Definition

For $A \subseteq X$, A is **universally null** if $\mathcal{N}(A) = \mathcal{M}_c(X)$.

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- universally null $\Longrightarrow$ residually null
- any perfect meager set is residually null but not universally null

Proposition (Extended BCT)

The collection of residually null subsets of a Polish space X is a σ-ideal. Moreover, this σ-ideal is proper, trivially, as $\mu(X) = 1$ for all $\mu \in \mathcal{M}(X)$.

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Question

For a Polish space X, the collection $\mathcal{I}$ of all sets A so that there is a meager set M and a universally null set U with $A = M \cup U$ is a proper σ-ideal. Then $\mathcal{I} \subseteq \mathbf{RN}(X)$. Is it the case that $\mathbf{RN}(X) = \mathcal{I}$?

Theorem (C.)

Let X and Y be Polish spaces and suppose that $A \subseteq X$ is a universally null set which is non-meager. Then $A \times Y \in \mathbf{RN}(X \times Y)$ but $A \times Y$ cannot be written as the union of a meager set with a universally null set.

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We offer a sketch of the proof.

Lemma

Let X and Y be Polish spaces. Then, for $\varepsilon > 0$, the set

$$\{\mu \in \mathcal{M}(X \times Y) : (\exists x \in X)(\mu(\{x\} \times Y) \geqslant \varepsilon)\}$$

is closed and nowhere dense.

Lemma

Let X and Y be Polish spaces and $A \subseteq X$ be universally null. Then, for $\mu \in \mathcal{M}(X \times Y)$,

$$\mu^*(A \times Y) > 0 \iff (\exists x \in A)(\mu(\{x\} \times Y) > 0).$$

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To finish the argument,

- Let A be universally null and non-meager
- $A \times Y$ is residually null
- By Kuratowski-Ulam Theorem, for any meager set M, $(A \times Y) \setminus M$ contains a perfect set of the form $\{x\} \times F$
- $(A \times Y) \setminus M$ is not universally null.

Extended Baire Property

Definition

We define $\mathrm{EBP}(X)$ to be all sets A so that there exists an open set U so that $U \triangle A := (A \cup U) \setminus (A \cap U)$ is residually null. Additionally, given a map $f : X \to Y$ where X and Y are Polish, f is said to be **EBP-measurable** if $f^{-1}[V] \in \mathrm{EBP}(X)$ for each open set $V \subseteq Y$.

As all meager sets are residually null, the class of $\mathrm{EBP}(X)$ contains all sets with the Baire property.

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As all meager sets are residually null, the class of $\mathrm{EBP}(X)$ contains all sets with the Baire property.

If sets which are residually null but non-meager exist, then the class $\mathrm{EBP}(X)$ is a finer class than the class of sets with the Baire property.

In a similar vein, we offer

Definition

Let X be a Polish space. We say that a set $A \subseteq X$ is

- **universally measurable** if A is μ-measurable with respect to all continuous measures or
- **residually measurable** if A is μ-measurable with respect to a co-meager collection of measures.

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Clearly, all universally measurable sets are residually measurable.

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Clearly, all universally measurable sets are residually measurable.

For any closed meager subset F, pick some continuous measure μ on F and let $A \subseteq F$ be a non-μ-measurable set. Then A is residually measurable but not universally measurable.

Definition

Let G be an abstract multiplicative group with a Polish topology τ. We say that G (paired with τ) is a **Polish group** provided the maps

- $g \mapsto g^{-1}$, $G \to G$, and
- $(g, h) \mapsto gh$, $G^2 \to G$

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Theorem (C., Kallman)

Let G and H be Polish groups and suppose $\phi : G \to H$ is a group homomorphism. If ϕ is EBP-measurable, ϕ is continuous.

σ-ideal	σ-algebra	continuity
meager	**BP**	✓
↓	↓	
RN	**EBP**	✓
↑		
UN		
	UM	???
	↓	
	RM	???

Martin's Axiom

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Suppose X is a compact Hausdorff space with countable cellularity and let $\mathrm{meag}(X)$ be collection of meager sets. Then $\mathrm{cov}(\mathrm{meag}) \geqslant \mathfrak{c}$.

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MA implies that, for a Polish space X and a cardinal $\kappa < \mathfrak{c}$,

- κ-length unions of meager sets are meager
- if $\mu \in \mathcal{M}_c(X)$, then κ-length unions of μ-null sets are μ-null

If we assume Martin's Axiom holds, we obtain

Theorem (C.)

Suppose X is Polish. For any cardinal $\kappa < \mathfrak{c}$ and any family $\mathcal{F}$ of κ-many residually null sets, $\bigcup \mathcal{F}$ is residually null.

Future Work

- X – compact Hausdorff space
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Residually Null Martin's Axiom (RNMA)

Suppose X is a compact Hausdorff space with countable cellularity and let $\mathbf{RN}(X)$ be collection of residually null sets. Then $\mathrm{cov}(\mathbf{RN}) \geqslant \mathfrak{c}$.

1. Is MA equivalent to RNMA?

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2. Does the residually null analog to the Kuratowski-Ulam Theorem hold?

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2. Does the residually null analog to the Kuratowski-Ulam Theorem hold?
3. Inspired by the construction of a residually null set which fails to be the union of a meager set and a universally null set in a space $X \times Y$, can such a set can be constructed in all Polish X?

Thank You!

[1] Lester Dubins and David Freedman, *Measurable sets of measures*, Pacific Journal of Mathematics **14** (1964), no. 4, 1211–1222.